

Seokbin Yoon

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RESEARCH INTERESTS

My research focuses on developing autonomous systems that are reliable, safe, and capable of high level reasoning. I am particularly interested in generative models for prediction and planning under uncertainty, and in methods that enable machines to perform structured decision making with high reliability. I believe machines surely can do what humans have done, such as air traffic control and driving, and at some point will perform these tasks even better.

EDUCATION

George Washington University

Doctor of Philosophy - PhD, Mechanical and Aerospace Engineering

Aug. 2026 – Present

Washington, DC, USA

- Advisor: Dr. Peng Wei
- Key Courses: Large Language Vision Models, Advanced Software Paradigms

Korea Aerospace University

Master of Science - MS, Air Transport, Transportation, and Logistics

March. 2022 – Feb. 2024

Goyang, South Korea

- Advisor: Dr. Keumjin Lee
- Key Courses: Advanced Computer Vision, Unmanned Air Transportation, Machine Learning, Airport Design with Simulation, Airport Planning, Probability and Statistics, Probability Models, Economic Analysis of Engineering Project

Korea Aerospace University

Bachelor of Science - BS, Air Transport, Transportation, and Logistics

March. 2020 – Feb. 2022

Goyang, South Korea

- Key Courses: Air Traffic Management, Aeronautical Telecommunications, Aviation Meteorology, Flight Operation Planning & Control, ATC Training, Aviation Weather Service, Air Navigation and Facility, Aeronautical Information Service, Air Transport System, Aviation Policy, Aviation Law

PUBLICATIONS

Journal Publications

- J1.** Seokbin Yoon, K. Lee, "Multi-Agent Inverted Transformer for Flight Trajectory Prediction," *IEEE Transactions on Intelligent Transportation Systems*, 2026, available at [\[here\]](#)
- J2.** Seokbin Yoon, K. Lee, "Aircraft Trajectory Dataset Augmentation in Latent Space," *International Journal of Aeronautical & Space Sciences*, 2025, available at [\[here\]](#)
- J3.** Seokbin Yoon, K. Lee, "Aircraft Trajectory Prediction with Inverted Transformer," *IEEE Access*, 2025, available at [\[here\]](#)

Conference Publications

- C1.** J. Lee, Seokbin Yoon, K. Lee, H. Baik, S. Jung, "Airport Terminal Passenger Queue Forecasting for Departure Gates and Security Checkpoints," *AIAA/IEEE Digital Avionics System Conference (DASC)*, 2026, available at [\[here\]](#)
- C2.** H. Kim, B. Kang, Seokbin Yoon, D. Jeon, K. Lee, "Reading the ATC Display: Vision Transformers Learn Air Traffic Complexity From Radar Imagery," *Air Transport Research Society World Conference*, 2026

- C3.** K. Kim, **Seokbin Yoon**, K. Lee, “Probabilistic Multi-Agent Aircraft Landing Time Prediction,” *AIAA SciTech*, 2026, , available at [\[here\]](#)
- C4.** H. Chai, **Seokbin Yoon**, K. Lee, “Learning to Explain Air Traffic Situation,” *US-Europe Air Transportation Research & Development Symposium*, 2025, available at [\[here\]](#)
- C5.** J. Kang, **Seokbin Yoon**, K. Lee, SH. Kim, “Urban Air Mobility Fleet Rebalancing with Real-Time Updates on Estimated Time of Arrival,” *AIAA/IEEE Digital Avionics System Conference (DASC)*, 2024, , available at [\[here\]](#)
- C6.** **Seokbin Yoon**, K. Lee, “Improving Aircraft Trajectory Prediction Accuracy with Over-sampling Technique,” *AIAA/IEEE Digital Avionics System Conference (DASC)*, 2023, available at [\[here\]](#)
- C7.** NH. Gray, S. Shin, **Seokbin Yoon**, Y. Park, H. Chai, K. Lee, H. Kim, S. Jung, “Safety and Capacity Analysis Framework for Integrated UAM Operation in Airports,” *AIAA/IEEE Digital Avionics System Conference (DASC)*, 2023, , available at [\[here\]](#)

HONORS AND AWARDS

- H1.** Best Student Paper at AIAA SciTech, 2026
- H2.** Best Paper of the Session at AIAA/IEEE Digital Avionics Systems Conference, 2024
- H3.** Best Paper of the Session at AIAA/IEEE Digital Avionics System Conference, 2023
- H4.** President’s Award, Korea Aerospace University, 2023
- H5.** Brain Korea 21 Plus (BK21+) Research Scholarship at Korean Government, 2022

CERTIFICATIONS AND LICENSES

- L1.** Air Traffic Controller, Ministry of Land, Infrastructure and Transport, South Korea, 2023
- L2.** Machine Learning with Python-From Linear Models to Deep Learning, edX, 2023

REVIEW SERVICES

Journal Reviews

- AIAA Journal of Aerospace Information Systems

TECHNICAL SKILLS

Programming : Python, MATLAB, $\text{\LaTeX}$

Frameworks : PyTorch, NumPy, Scikit-learn, OpenCV

Languages : Korean (Native), English